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Week 1 / Quiz 1

An Operating System controls only hardware of a computing device.

T

True

F

False

SUBMIT ANSWER

2 of 10

Part of an operating system that performs essential tasks to manage a computer system is called a _____.

A

Virtual Machine

B

MBR

C

bootloader

D

kernel

E

GRUB

SUBMIT ANSWER

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The general name of a program that is used to load an operating system kernel is called a _____.

A

Virtual Machine

B

MBR

C

bootloader

D

kernel

E

GRUB

SUBMIT ANSWER

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An example of program stated in the previous question that is used by many Linux operating systems is _____.

A

Virtual Machine

B

MBR

C

bootloader

D

kernel

E

GRUB

SUBMIT ANSWER

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The program in the previous question is stored on the _____.

A

Virtual Machine

B

MBR

C

bootloader

D

kernel

E

GRUB

SUBMIT ANSWER

Concept of virtual machines can be applied to programming languages. (True)

Concept of virtual machines can be applied to cloud computing (True)

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Parts of memory spaces are preserved for OS.

☒ True ☐ False

Remix

SUBMIT ANSWER

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Install two OSes in different partitions of a hard disk is considered using virtual machine concept.

☐ True ☒ False

SUBMIT ANSWER

8 of 10

An OS that runs on a virtual machine is called _____.

☐ A Virtual OS ☒ B Guest OS
☐ C Host OS ☐ D Embedded OS
☐ E Realtime OS

SUBMIT ANSWER

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You cannot install the same type of OS (e.g. Windows 10 and Windows 10) on the real machine and virtual machine.

☐ True ☒ False

SUBMIT ANSWER

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JVM is a Java Runtime.

☒ True ☐ False

SUBMIT ANSWER

When we say “ computing resources”, we mean only hardware. (F)

Os can manage a computer user. (T)

The program that is used to load a program in the boot sector is stored on the _____. (MBR)

The first program that will be loaded and run when we turn on a computer is _____. (UEFI)

The BIOS program is stored in a computer’s _____. (ROM)

The instructions that can be used only by OS are a group of instructions called _____. (Privileged instructions)

The instructions to read the current system date and time is not necessary to be privileged. (T)

The instructions to read the secure OS information should be privileged. (True)

Os that controls the physical hardware is called _____. (host OS)

Oracle virtual box is an example of a _____. (Virtualization software)

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MULTIPLE CHOICE

FINISH QUIZ

Week 2

Device controller informs CPU that it has finished its operation by causing a(n) _____.

- | | |
|--|--|
| ☐ A Device Driver | ☒ B Interrupt |
| ☐ C Controller | ☐ D System Call |
| ☐ E System Program | |

2 of 11

The program that is used to handle the thing in the previous question is called _____.

- | | |
|--|--|
| ☐ A Device Driver | ☐ B Interrupt vector |
| ☒ C Interrupt Service Routine | ☐ D System Call |
| ☐ E System Program | |

SUBMIT ANSWER

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The program in the previous question is searched through the thing called _____.

- | | |
|---|---|
| ☐ A Device Driver | ☒ B Interrupt vector |
| ☐ C Interrupt Service Routine | ☐ D System Call |
| ☐ E System Program | |

SUBMIT ANSWER

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The System that allows only programs that work without user interaction is called _____.

- | | |
|---|--|
| ☒ A Batch System | ☐ B Interactive system |
| ☐ C Hybrid System | ☐ D Parallel System |
| ☐ E Distributed System | |

SUBMIT ANSWER

5 of 11

Time Sharing in an interactive system is the sharing memory among programs.

- | | |
|------------------------------|--|
| ☐ T True | ☒ F False |
|------------------------------|--|

SUBMIT ANSWER

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TRUE / FALSE

FINISH QUIZ

In a hybrid system batch programs are running in the background mode.

- | | |
|---|-------------------------------|
| ☒ T True | ☐ F False |
|---|-------------------------------|

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Windows 10 is an example of _____ operating systems.

- | | |
|--|--|
| ☐ A Single-user single-tasking | ☒ B Single-user multi-tasking |
| ☐ C multi-user multi-tasking | ☐ D Single Program Batch |
| ☐ E Multi-Program Batch | |

SUBMIT ANSWER

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Time critical system.

- | | |
|---|---|
| ☐ A Batch System | ☐ B Time Sharing System |
| ☒ C Real Time System | ☐ D Distributed System |
| ☐ E Parallel System | |

SUBMIT ANSWER

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Multiprocessors system

- | | |
|--|---|
| ☐ A Batch System | ☐ B Time Sharing System |
| ☐ C Real Time System | ☐ D Distributed System |
| ☒ E Parallel System | |

SUBMIT ANSWER

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Loosely coupled system

- | | |
|--|---|
| ☐ A Batch System | ☐ B Time Sharing System |
| ☐ C Real Time System | ☒ D Distributed System |
| ☐ E Parallel System | |

SUBMIT ANSWER

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For distributed OS, resources are owned by local nodes in the network.

- | | |
|---|-------------------------------|
| ☒ T True | ☐ F False |
|---|-------------------------------|

SUBMIT ANSWER

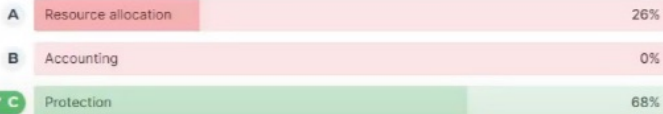
1. The soft real-time system requires a specific operating system. (F)
2. Any server OS is an example of _____ operating systems. (multi-user multi-tasking)
3. MS DOS is an example of _____ operating systems. (single-user single-threading)

2. OS service that involves ensuring that all access to system resources is controlled.

Week 4 (26-7)

Hide Results

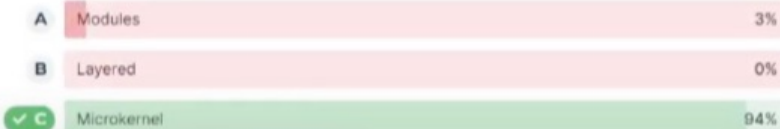
34/43 Students Answered



11. Kernel architecture that focuses to keep a kernel to have small size is _____ architecture.

Hide Results

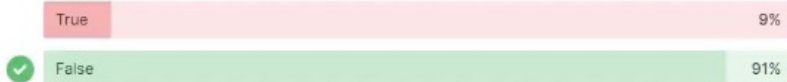
34/43 Students Answered



13. The benefit of using high-level API is that it makes the program run faster.

Hide Results

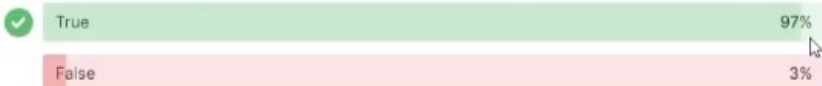
34/43 Students Answered



9. OS handles system call activation similar to handle a hardware interrupt.

Hide Results

34/43 Students Answered



6. The "cp" command is a command built-in in a Unix-based OS.

Show Results

34/43 Students Answered

True

False

14. Command prompt in MS Windows is an example of a CLI program.

[Show Results](#)

34/43 Students Answered

True

100%

10. Our program never uses system calls if we do not call a system call directly.

[Hide Results](#)

34/43 Students Answered

True

0%



False

100%

1. Part of OS that provides interfaces for applications that need to use OS services.

[Hide Results](#)

34/43 Students Answered

A

System programs

9%



B

System Calls

71%

C

Driver library

0%

15. Bash is an example of a CLI program.

[Hide Results](#)

34/43 Students Answered



True

100%

False

0%

8. From the following Unix commands, a command built-in is ____.

[Hide Results](#)

34/43 Students Answered

A

ls

3%

B

mv

0%

C

rm

0%



D

cd

97%

12. Kernel architecture that focuses on performance is _____ architecture.

[Hide Results](#)

34/43 Students Answered

A	Modules	0%
B	Layered	3%
C	Microkernel	3%
✓ D	Monolithic	94%

7. Implementing a command as a command built-in makes the command easy to modify.

[Hide Results](#)

34/43 Students Answered

	True	44%
✓	False	56%

4. Common communication service provided by OS

[Hide Results](#)

34/43 Students Answered

A	resource allocations	3%
B	program execution	0%
✓ C	message passing	94%
D	accounting	3%

5. Command Line Interface (CLI) cannot be implemented in a kernel

[Hide Results](#)

34/43 Students Answered

	True	15%
✓	False	85%

3. OS service that involves authentication.

[Hide Results](#)

34/43 Students Answered

A	Resource allocation	0%
B	Accounting	0%
C	Protection	0%
✓ D	Security	100%

1. A process that waits for an OS to choose it to run is in the state called _____ state.

Hide Results

31/43 Students Answered

A	new	10%
✓ B	ready	90%
C	running	0%

2. A process is running, but when it needs to do an I/O operation its state will be changed to _____

Hide Results

31/43 Students Answered

A	new	0%
B	ready	0%
C	running	0%
✓ D	waiting	100%
E	terminated	0%

3. Which of the following state transitions cannot be occurred?

Hide Results

31/43 Students Answered

A	Ready → Running	0%
B	running → terminated	3%
C	running → waiting	0%
✓ D	waiting → running	97%
E	waiting → ready	0%

[Back to Results Table](#)

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4. Process information is stored in a _____.

Hide Results

31/43 Students Answered

A	Job queue	0%
B	Ready queue	0%
C	Device queue	0%
✓ D	PCB	74%
E	Register	26%

5. Switching a CPU to another process is called _____.

Hide Results

31/43 Students Answered

A	Program Counter	0%
B	Process Number	0%
✓ C	Context Switch	100%
D	Dispatcher	0%

< Back to Results Table

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6. Part of OS that does the context switch is _____.

Hide Results

31/43 Students Answered

A	Program Counter	3%
B	Process Number	0%
C	Context Switch	0%
✓ D	Dispatcher	84%
E	Scheduler	13%

7. Part of OS that is responsible for choosing a process to be executed by a CPU is _____.

Hide Results

31/43 Students Answered

A	Program Counter	3%
B	Process Number	0%
C	Context Switch	6%
D	Dispatcher	3%
✓ E	Scheduler	87%

8. Consider the program in the picture.
How many processes will be created (including the parent process)?

Hide Results

31/43 Students Answered

A	1	3%
B	2	6%
✓ C	3	66%
D	4	23%

```
1  /* forkExample.c */
2  #include <stdio.h>
3  #include <stdlib.h>
4  #include <unistd.h>
5  #include <sys/types.h>
6  #include <sys/wait.h>
7  int main()
8  {
9      pid_t pid;
10     int i;
11     for (i = 0; i < 2; i++) {
12         pid = fork();
13         if (pid == 0) {
14             printf("child %d\n", i+1);
15             exit(0);
16         }
17     }
18     while(wait(NULL) != -1);
19     printf("good bye\n");
20     return 0;

```

9. Consider the program in the picture.
What is (are) the possible output(s)?

Hide Results

31/43 Students Answered

A	child 1 goodbye child 2	13%
B	child 2 child 1 goodbye	81%
C	goodbye child 2 child 1	0%

```
1  /* forkExample.c */
2  #include <stdio.h>
3  #include <stdlib.h>
4  #include <unistd.h>
5  #include <sys/types.h>
6  #include <sys/wait.h>
7  int main()
8  {
9      pid_t pid;
10     int i;
11     for (i = 0; i < 2; i++) {
12         pid = fork();
13         if (pid == 0) {
14             printf("child %d\n", i+1);
15             exit(0);
16         }
17     }
18     while(wait(NULL) != -1);
19     printf("goodbye\n");
20     return 0;
21 }
```

Back to Results Table

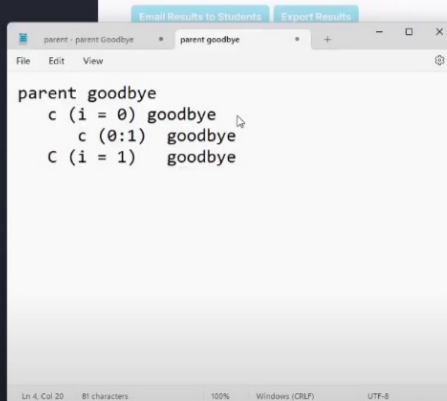
<

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>

10. Consider the program in the picture.
How many times that the word "goodbye" will be printed?

```
1  /* forkExample.c */
2  #include <stdio.h>
3  #include <stdlib.h>
4  #include <unistd.h>
5  #include <sys/types.h>
6  #include <sys/wait.h>
7  int main()
8  {
9      pid_t pid;
10     int i;
11     for (i = 0; i < 2; i++) {
12         pid = fork();
13         if (pid == 0) {
14             printf("child %d\n",
15                 i+1);
16             exit(0);
17         }
18     }
19     while(wait(NULL) != -1);
20     printf("goodbye\n");
21     return 0;
22 }
```



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Quiz 3

Part of OS that provides interfaces for applications that need to use OS services.

- | | |
|---|---|
| ☐ A System programs | ☒ B System Calls |
| ☐ C Program library | ☐ D User Interface |
| ☐ E batch | |

SUBMIT ANSWER

2 of 8

OS service that involves ensuring that all access to system resources is controlled.

- | | |
|--|------------------------------------|
| ☒ A Resource allocation | ☐ B Accounting |
| ☐ C Protection | ☐ D Security |
| ☐ E Error Detection | |

SUBMIT ANSWER

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Common communication services provided by OS

- | | |
|--|---|
| ☐ A resource allocations | ☐ B program execution |
| ☐ C message passing | ☐ D shared memory |
| ☒ E C and D | |

SUBMIT ANSWER

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Command Line Interface (CLI) cannot be implemented in a kernel

- | | |
|------------------------------|--|
| ☐ T True | ☒ F False |
|------------------------------|--|

SUBMIT ANSWER

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whereis [command]
is an example of command built-in. type [command]

- | | |
|------------------------------|--|
| ☐ T True | ☒ F False |
|------------------------------|--|

SUBMIT ANSWER

6 of 8

OS handles system call activation similar to handle a hardware interrupt.

- | | |
|---|-------------------------------|
| ☒ T True | ☐ F False |
|---|-------------------------------|

SUBMIT ANSWER

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If we do not call to a system call directly, this means that our program never uses system calls.

- | | |
|------------------------------|--|
| ☐ T True | ☒ F False |
|------------------------------|--|

SUBMIT ANSWER

8 of 8

Kernel architecture that focuses to keep a kernel to have small size is _____ architecture.

- | | |
|--|------------------------------------|
| ☐ A Modules | ☐ B Layered |
| ☒ C Microkernel | ☐ D Monolithic |
| ☐ E none of the above. | |

SUBMIT ANSWER